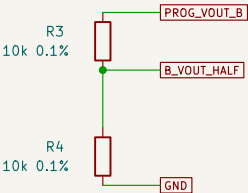
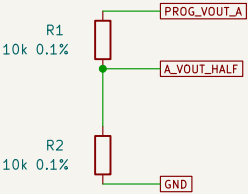
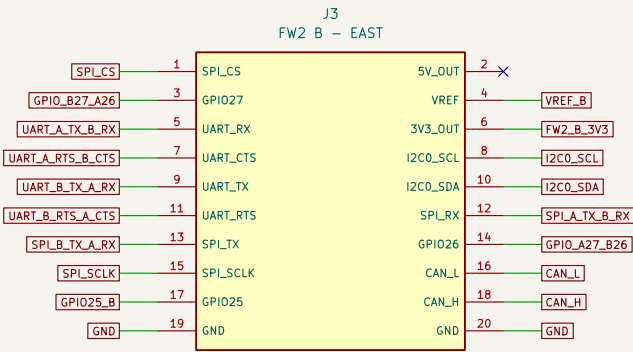
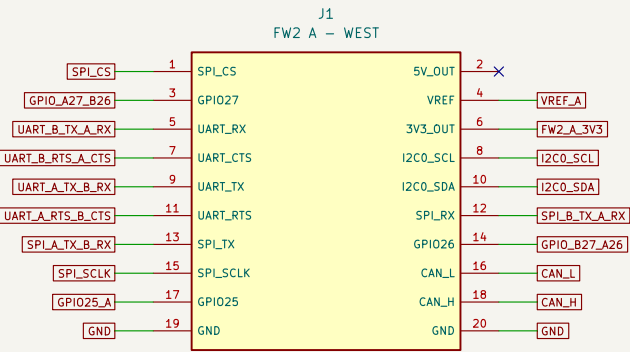


GPIO JOINER / REV A / TWO-DEVICE TEST FIXTURE

SIDE A: choose WEST FW2 or NORTH OG

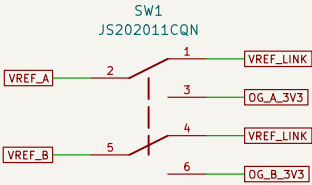
SIDE B: choose EAST FW2 or SOUTH OG

DIVIDED PROGRAMMABLE OUTPUTS



Firmware: VOUT = 2 x opposite AIN2
0.1% divider; nominal range 0 to 2.75 V.

VREF MODE – SWITCH WITH POWER OFF

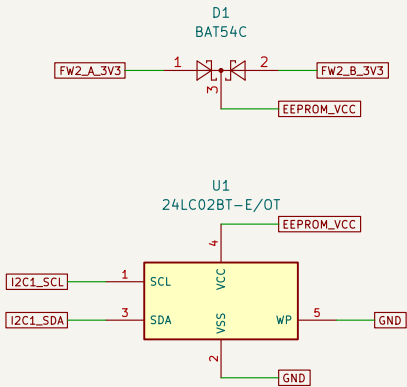


FW2/mixed: contacts 1–2 and 4–5; linked VREF.
Enable exactly ONE FW2 IO voltage source.
OG/OG: contacts 2–3 and 5–6; local 3.3 V.

ONE DEVICE PER SIDE / TWO DEVICES MAXIMUM

Named nets implement SPI, UART and GPIO swaps. I2C0/I2C1 are direct.
5 V outputs and OG SWD are intentionally not connected.

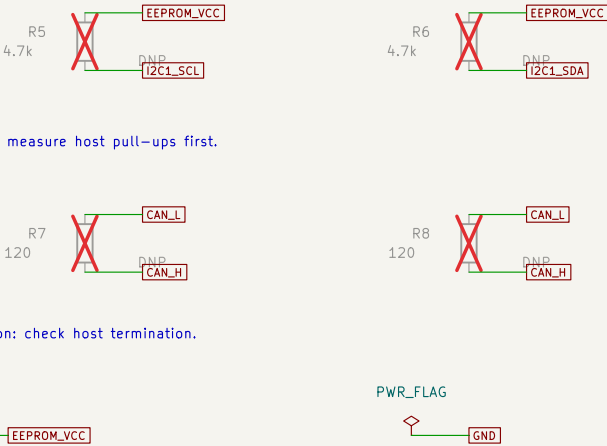
ORCA IDENTIFICATION / SHARED I2C1



U1 WP tied low for programming. EEPROM at 0x50.
BAT54C isolates the two FW2 3.3 V outputs.
EEPROM image and host power/pull-ups need bench validation.

PROTOTYPE – VERIFY MATING AND FIRMWARE BEFORE FABRICATION

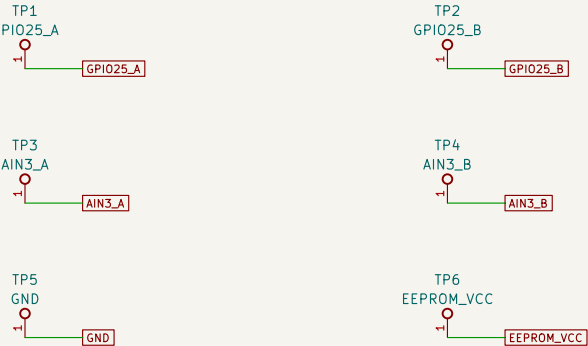
OPTIONAL BUS COMPONENTS



DNP pull-ups: measure host pull-ups first.

DNP termination: check host termination.

SPARE SIGNALS / BENCH TEST POINTS



Sheet: /
File: gpiojoiner.kicad_sch

Title: GPIO Joiner – two-device GPIO test fixture

Size: A3 Date: 2026-09-08

KiCad E.D.A. 10.0.6

Rev: A

Id: 1/1